

CSE331: Automata and Computability

Assignment 3

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Show the following language is not regular using pumping lemma

1. $L = \{ 0^{2n}1^n \mid n \geq 0 \}$, where $\Sigma = \{0, 1\}$.

- Assuming : Suppose L is a regular language, then it must satisfy the pumping property.
- Choosing a string : $w = 0^{2s}1^s$ $[2s + s \geq s]$
- Splitting : $w = x \cdot y \cdot z$ [where $|xy| \leq s$ and $|y| \geq 1$]
 $x = 0^p$; $y = 0^q$; $z = 0^{s-p-q}0^s1^s$
- Checking if xy^iz belongs to L for all $i \geq 0$:
 $i=0$: $xy^0z = xz = 0^p \cdot 0^{s-p-q} \cdot 0^s \cdot 1^s = \cancel{0^p} \cdot \cancel{0^s} \cdot \cancel{0^p} \cdot 0^s \cdot 1^s = 0^{2s-q} \cdot 1^s$
Here, 0 occurs $2s-q$ times and 1 occurs s times. But 0 should occur $2n$ times (twice of 1s occurrences). So, $0^{2s-q} \cdot 1^s \notin L$, a contradiction. L is not regular.

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2. $L = \{ w \text{ is not a palindrome} \}$, where $\Sigma = \{0, 1\}$.

Solⁿ:

If a language is not regular, its complement is also not regular. So we will first prove that $\bar{L}$ is not regular.

$$\bar{L} = \{ w \text{ is a palindrome} \}, \text{ where } \Sigma = \{0, 1\}$$

- Suppose $\bar{L}$ is regular. Then it must satisfy the Pumping property.

- $w = 0^s 1 0^s$ where $|w| \geq s$
 $2s+1 \geq s$

- $w = xyz$ where $|xy| \leq s$ and $|y| \geq 1$

$$\begin{array}{ccc} \swarrow & \downarrow & \downarrow \\ 0^p & 0^q & 0^{s-p-q} 1 0^s \end{array}$$

- Also $xy^p z$ must belong to $\bar{L}$ for all $p \geq 0$.

$$\begin{aligned} p=2, xy^2 z &= 0^p 0^q 0^q 0^{s-p-q} 1 0^s \\ &= \cancel{0^p} \cancel{0^q} 0^q 0^s \cancel{0^q} \cancel{0^q} 1 0^s \\ &= 0^q 0^s 1 0^s \\ &= 0^{q+s} 1 0^s \notin L \end{aligned}$$

- Contradiction! Hence L is not regular.

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3. $L = \{a^n \text{ where } n \text{ is prime}\}$, where $\Sigma = \{a\}$.

primes : $\{2, 3, 5, 7, 11, \dots\}$

$L : \{aa, aaa, aaaaa, \dots\}$

• Assuming : L is a regular language and satisfies the pumping property.

• Choosing a string : $w = a^s$ [$s \geq 5$]

• Splitting : $w = xyz$ [where $|xy| \leq s$ and $|y| \geq 1$]

$$x = a^p, y = a^q, z = a^{s-p-q}$$

• checking if $xy^iz \in L$ for all $i \geq 0$:

$$i=0 : xy^0z = xz = a^p \cdot a^{s-p-q} = a^{s-q} ?$$

$$i=1 : xy^1z = xyz = a^p \cdot a^q \cdot a^{s-p-q} = a^s ?$$

$$i=s+1 : xy^{(s+1)}z = xy^s \cdot y^1 \cdot z = a^p \cdot a^{sq} \cdot a^q \cdot a^{s-p-q} = a^{sq+s} = a^{s(s+1)} \notin L$$

violates $|y| \geq 1$
 $2 \times (0+1)$
 $3 \times (0+1)$

Here, the length of w is $s \times (q+1)$. But a prime number is only divisible by 1 and itself. The length of w is divisible by s and $(q+1)$, making it non-prime except for when $s = \text{prime}$ and $q=0$, but q cannot be zero according to the condition in step 3, $|y| \geq 1$. Thus L is not regular.

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4. $L = \{a^m b^n \mid m \neq n\}$, where $\Sigma = \{a, b\}$.

• Assuming : that L is regular and satisfies the pumping property

• Choosing w : $w = a^s b^{s+1}$ [$s + s + 1 \geq s$]

• Splitting : $w = xyz$ [$|xy| \leq s$ and $|y| \geq 1$]

$$x = a^p, \quad y = a^q, \quad z = a^{s-p-q} b^{s+1}$$

• Checking if $xy^i z \in L$ for all $i \geq 0$:

• $i=1$: $xyz = \cancel{a^p} \cancel{a^q} a^s \cancel{a^{-p}} \cancel{a^{-q}} b^{s+1} = a^s b^{s+1}$

• $i=2$: $xyyz = \cancel{a^p} \cancel{a^q} \cancel{a^q} a^s \cancel{a^{-p}} \cancel{a^{-q}} b^{s+1} = a^{s+q} b^{s+1}$

• $i = \frac{s+1}{q} + 1$: $xy^{\left(\frac{s+1}{q} + 1\right)} z = a^p (a^q)^{\left(\frac{s+1}{q} + 1\right)} a^{s-p-q} b^{s+1}$
 $= a^p a^{(s+1)+q} a^s a^{-p} a^{-q} b^{s+1}$
 $= \cancel{a^p} a^{s+1} \cancel{a^q} a^s \cancel{a^{-p}} \cancel{a^{-q}} b^{s+1}$
 $= a^{s+1} b^{s+1} \notin L$

Contradiction, when $i = \frac{s+1}{q} + 1$, w has equal number of a 's and b 's.
 L is not a regular language.

Context free grammar

1. $L = \{w \in \{0,1\}^* : w = 0^n 1^n, \text{ where } n \text{ is odd}\}$

$$\begin{aligned} S &\rightarrow 0P1 \\ P &\rightarrow 0S1 \mid \epsilon \end{aligned}$$

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2. $L = \{ w \in \{0,1\}^* : \text{the length of } w \text{ is divisible by } 3 \}$

$$S \rightarrow psp | \epsilon$$

$$p \rightarrow 0 | 1$$

3. $L = \{ w \in \{0,1\}^* : w_1 \# w_2, \text{ the number of 0's in } w_1 \text{ is equal to number of 1's in } w_2 \}$

$$S \rightarrow 1S | S0 | 0S1 | \#$$

4. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i = k, i, k \geq 1 \text{ and } j \geq 2 \}$

$$w = 0^i 2^j 1^k$$

$$= 0^k 2^j 1^k$$

$$S \rightarrow 0P1$$

$$P \rightarrow 0P1 | y$$

$$y \rightarrow 22x$$

$$x \rightarrow 2x | \epsilon$$

$i, k \geq 1$
 $j \geq 2$

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5. $L = \{ w \in \{0,1,2\}^* : w = 0^i 2^j 1^k, \text{ where } i \text{ is multiple of two, } k \text{ is two more than multiple of 3, } j = k+i \text{ and } i,j,k \geq 0 \}$

$$\begin{aligned}
 w &= 0^i 2^j 1^k \\
 &= 0^{2i} 2^{3k+i+2} 1^{3k+2} \\
 &= \underbrace{0^{2i} 2^i}_{x} \underbrace{2^k 1^3}_{y} 1^2
 \end{aligned}$$

$$\begin{aligned}
 j &= k+i \\
 &= 3k+2+2i \\
 &= 2i+3k+2
 \end{aligned}$$

$$S \rightarrow xy$$

$$x \rightarrow 00x22 \mid 22$$

$$y \rightarrow 222y111 \mid 11$$

6. $L = \{ w \in \{0,1\}^* : \text{the number of 0s and 1s are different in } w \}$

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$L_7 = \{ w \in \{0,1\}^* : \text{The number of 0's and 1's different in } w \}$

First Solve the CFG for equal 0's and 1's

$S \rightarrow OS1 \mid 1SO \mid \underline{SS} \mid \epsilon$

$\underbrace{\hspace{1.5cm}}_{\text{For handling } 0110}$

A B

0110 110010

Both blocks contain equal amount of 0's and 1's. To make it unequal I can insert 0's before block A, in middle of A and B and after block B.

$T \rightarrow SOS$

$S \rightarrow OS1 \mid 1SO \mid \underline{SS} \mid OS \mid \epsilon$

There could also be more 1's than 0's.

$S \rightarrow A \mid B$

$A \rightarrow XOX$

$X \rightarrow OX1 \mid 1X0 \mid XX \mid OX \mid \epsilon$

$B \rightarrow J1J$

$J \rightarrow OJ1 \mid 1J0 \mid JJ \mid 1J \mid \epsilon$

Regular expression to CFG

$(a^*b^* + (ac + b^*c)b)^*$

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$$\underbrace{\underbrace{a^*b}_{X} + \underbrace{(ac + b^*c)b}_{Y}}_{Y}^*$$

$$S \rightarrow RS | \epsilon$$

$$R \rightarrow X | Y$$

$$X \rightarrow PQ$$

$$P \rightarrow aP | \epsilon$$

$$Q \rightarrow b$$

$$Y \rightarrow CD$$

$$D \rightarrow b$$

$$C \rightarrow E | F$$

$$E \rightarrow ac$$

$$F \rightarrow GH$$

$$H \rightarrow c$$

$$G \rightarrow bG | \epsilon$$

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Derivation, Parse tree and Ambiguity

Let $\Sigma = \{0, 1\}$. Consider the following grammar over Σ .

$$S \rightarrow 0S0 \mid 1S1 \mid A$$

$$A \rightarrow 00A \mid 01A \mid 10A \mid 11A \mid \varepsilon$$

- (a) **Give** a leftmost derivation for the string 01101010. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the given string in (a). (4 points)
- (d) How many four-letter strings will have exactly one parse tree in the given grammar? (1 point)

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$\begin{array}{c} S \\ \\ A \\ \backslash \\ 01A \\ \\ 10A \\ \\ 10A \\ \\ 10A \\ \\ \epsilon \end{array}$	$\begin{array}{l} S \\ \Rightarrow A \\ \Rightarrow 01A \\ \Rightarrow 0110A \\ \Rightarrow 011010A \\ \Rightarrow 01101010A \\ \Rightarrow 01101010 \end{array}$
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S 0 S 0 A 1 1 A 0 1 A 0 1 A ε	S ⇒ 0S0 ⇒ 0A0 ⇒ 011A0 ⇒ 01101A0 ⇒ 0110101A0 ⇒ 01101010
S 0 S 0 1 S 1 A 1 0 A 1 0 A ε	S ⇒ 0S0 ⇒ 01S10 ⇒ 01A10 ⇒ 0110A10 ⇒ 011010A10 ⇒ 01101010

Question (d): **8 strings**

(0001, 0011, 0101, 0111, 1000, 1010, 1100, 1110) → Students don't need to write these strings

You need to write only 1 string.